

20.1: Rigid Bodies:

A moment is defined from a pivot by the product of the magnitude of the force with the orthogonal distance between the pivot & line of action of the force.

Moments can be clockwise (negative) or anticlockwise (positive).

An object is in equilibrium if:

$\Rightarrow \sum_i F = 0$ in all planes.
$\Rightarrow \sum m_{\circlearrowright} = \sum m_{\circlearrowleft}$ / $\sum m = 0$ about any point.

Moments should be taken from an appropriate pivot.

A couple is 2 equal, but opposite forces acting on an object. The forces are parallel, and along different lines. This means $\sum F = 0$, but $\sum m = Fd$

This causes rotational motion, and no translational motion.